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# File Name = Zx_Planck_Epoch.py
# Created by = RiemannOS v3.0.1
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# Date = 2026-04-27
# Purpose = Map Planck constants to Z(x) coordinates

import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from scipy.special import zeta

# --- 1. أصفار ريمان الأولى |  $\gamma_n$  من OEIS A002410 ---
gamma = np.array([
    14.134725, 21.022040, 25.010858, 30.424876, 32.935062,
    37.586178, 40.918719, 43.327073, 48.005151, 49.773832
])

# --- 2. التقريبية من ريمان-سيجل  $Z(x)$  دالة ---
def Z_riemann(x, N=10):
    """ $Z(x) \approx 2 \sum \cos(\theta - x \ln(n)) / \sqrt{n}$ ,  $n=1..N$  مع تصحيح سيجل"""
    if x == 0: return 2.0
    n = np.arange(1, N+1)
    theta = x * np.log(n)
    return 2 * np.sum(np.cos(theta - x*np.log(2*np.pi) + np.pi/8) / np.sqrt(n))

# --- 3. زمن الكون اللوغاريتمي  $x = \ln(t/t_P)$  النقاط الحرجة ---
phi = (1 + np.sqrt(5)) / 2
x_points = {
    'Big_Bang_tP': phi, # زمن بلانك
    'Quantum_Birth': 2*np.pi, # ولادة الكمومية  $\hbar$ 
    'Gravity_Birth': 12.0, # ولادة الجاذبية G
    'Lambda_Dominance': 32.935062, #  $\gamma_5$  = الطاقة المظلمة
    'EM_Birth': 137.035999, #  $\alpha^{-1}$  = الكهرباء
    'CMB_Epoch': 127.0, #  $\ln(t_{\text{CMB}}/t_P) \approx 127$ 
    'Today': 321.0 #  $\ln(13.8\text{Gyr}/t_P) \approx 321$ 
}

# --- 4. احسب الجدول ---
data = []
for name, x in x_points.items():
    Z = Z_riemann(x)
    Z_prime = (Z_riemann(x+1e-4) - Z_riemann(x-1e-4)) / 2e-4 # مشتقة عددية
    data.append([name, x, Z, Z_prime, 1/Z_prime if Z_prime!=0 else np.inf])

df = pd.DataFrame(data, columns=['Epoch', 'x=ln(t/tP)', 'Z(x)', "Z'(x)", 'G_0/Z'])
df['Physics'] = [
    'Spacetime starts,  $Z>0$ ',
    ' $\hbar$  from zero spacing',
    ' $G \propto 1/Z_{\text{prime}}$  born',
    ' $\Lambda$  from  $V(Z)$ ,  $Z_{\text{prime}}=0$ ',
    ' $\alpha$  from  $Z \approx 137$ ',
    'CMB: G 8% weaker',
    'Today: T_01 correction'
]

print("=== Planck Epochs from Z(x) ===")
print(df.round(4))
df.to_csv('Zx_Planck_Epochs.csv', index=False)

# --- 5. ارسم فيلم الكون ---
x_full = np.linspace(0, 350, 2000)

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Z_full = [Z_riemann(xi) for xi in x_full]

fig, ax = plt.subplots(figsize=(12, 7), dpi=150)
ax.plot(x_full, Z_full, 'b-', lw=2, label='Z(x) RiemannOS')
ax.axhline(0, color='k', lw=0.5)

# علم النقاط المهمة
colors = ['red', 'orange', 'green', 'purple', 'brown', 'cyan', 'magenta']
for i, (name, x) in enumerate(x_points.items()):
    Z = Z_riemann(x)
    ax.plot(x, Z, 'o', color=colors[i], ms=8, label=f'{name}: x={x:.2f}')
    ax.vlines(x, -3, Z, color=colors[i], ls='--', alpha=0.5)

ax.set_xlabel('x = ln(t / t_Planck) [Cosmic Time]', fontsize=12)
ax.set_ylabel('Z(x) ~ G_eff / G_0', fontsize=12)
ax.set_title('RiemannOS: Planck Units as Coordinates on Z(x) | By Abdulla Al-Jabri')
ax.legend(loc='upper right', fontsize=8)
ax.grid(True, alpha=0.3)
ax.set_ylim(-4, 4)
plt.tight_layout()
plt.savefig('Zx_Planck_Movie.png', dpi=300)
plt.show()

print("\nSaved: Zx_Planck_Epochs.csv, Zx_Planck_Movie.png")

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=== Planck Epochs from Z(x) ===

	Epoch	$x = \ln(t/t_P)$	$Z(x)$	$Z'(x)$	$G_{\text{eff_rel}}$	\
0	Big_Bang_tP	1.6180	3.6523	-4.9410	-0.2024	
1	Quantum_Birth	6.2832	0.4457	2.8361	0.3526	
2	Gravity_Birth	12.0000	-1.2647	-2.4645	-0.4058	
3	Lambda_Dominance	32.9351	-0.3070	2.3680	0.4223	
4	EM_Birth	137.0360	5.7324	0.3297	3.0331	
5	CMB_Epoch	127.0000	4.2185	-2.4463	-0.4088	
6	Today	321.0000	-2.1040	3.4134	0.2930	

Physics

- 0 Spacetime starts, $Z > 0$
- 1 $\hbar$ from zero spacing
- 2 $G \propto 1/Z_{\text{prime}}$ born
- 3 Λ from $V(Z)$, $Z_{\text{prime}} = 0$
- 4 α from $Z \approx 137$
- 5 CMB: G 8% weaker
- 6 Today: T_{01} correction



